MODULE 6 · RECOVERY

Lesson 19: Upset Recovery Principles

Lesson 19 of 24 · about 8 min

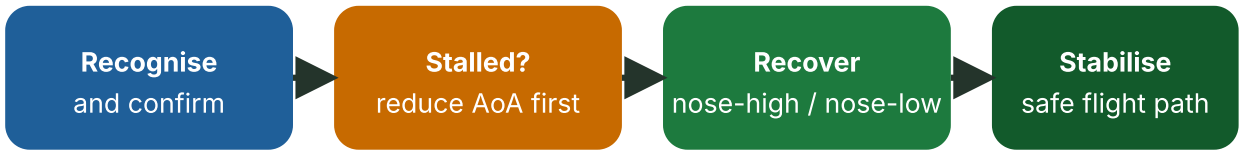
LEARNING OBJECTIVES

- State the general principles of upset recovery.
- Explain why stall recovery takes priority.
- Describe the correct use of automation, thrust, rudder and g during recovery.



Recognise, deal with any stall, recover, stabilise.

The recovery sequence



Throughout: communicate (PF/PM callouts), avoid excessive g, and follow the manufacturer's procedures.

Stall recovery always takes priority: you cannot recover attitude while the wing is stalled.

1. **Recognise and confirm** the situation — attitude first, cross-checked (Lesson 17).
2. **Disconnect the autopilot and autothrust** — they may be causing or worsening the upset, and their behaviour is unpredictable at extreme attitudes.

3. **If stalled, recover from the stall first** — reduce angle of attack (Lesson 20). Attitude cannot be recovered while the wing is stalled.
4. **Recover the attitude** using the nose-high or nose-low technique (Lessons 21 and 22).
5. **Return to a safe flight path** — level flight or a climb, appropriate speed, trimmed.

General principles

- **Use the manufacturer's procedures.** The generic principles here explain *why*; the FCOM and QRH tell you *how* for your type.
- **Pitch is controlled with elevator and stabiliser trim.** Use trim carefully to assist a large pitch input, not as the primary control.
- **Roll with aileron and spoiler.** Full deflection may be needed. Rudder is normally not required and must never be used in rapid alternating inputs.
- **Thrust is used as required** — remember pitch-power coupling: reducing thrust can help lower the nose, adding thrust helps a nose-low recovery only after the wings are level and the speed allows it.
- **Respect g limits.** Unload in a nose-high recovery; avoid a rolling pull and excessive g in a nose-low recovery.
- **Do not change configuration** (flaps, gear) unless the procedure requires it. Speedbrakes may be used in high-speed nose-low situations.
- **Communicate** — announce the upset and the recovery; the pilot monitoring calls out attitude, speed and altitude.

Common errors

- Pulling back in a stall or a nose-low, high-bank attitude.
- Applying full thrust in a nose-high, low-speed upset on an aircraft with underwing engines.
- Over-controlling — large inputs in both directions, especially at high altitude.
- Using rudder to roll, or making alternating rudder inputs.
- Failing to disconnect the automation, or re-engaging it too early.

KEY TAKEAWAYS

- Recognise and confirm → automation off → stall recovery first → attitude recovery → safe flight path.
- Elevator and trim for pitch; aileron and spoiler for roll; rudder normally not needed.
- Use thrust as required, with pitch-power coupling in mind.
- Communicate throughout; follow the manufacturer's procedures.

KNOWLEDGE CHECK

An aircraft is nose-high, rolling, with the stall warning active. What is the priority?

A. Roll the wings level first

B. Recover from the stall — reduce the angle of attack

C. Apply full thrust

D. Extend the flaps

Answer: B. Stall recovery always takes priority. Attitude cannot be recovered while the wing is stalled.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.